

Application form mySNF

Instrument **Scientific Exchanges**

Part 1: General Information

Basic data

Project Title	Biodiversity Dynamics in Coevolutionary Metaecosystems (research visit)
Project title in English	Biodiversity Dynamics in Coevolutionary Metaecosystems (research visit)

Research Field	Life sciences
Main Discipline	30207 Ecology
University	Eidg. Anstalt für Wasserversorgung - EAWAG

Applicant(s)

Main Applicant	Carlos Melian
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Grant Application

Amount requested (CHF)	Total	7'500
Requested starting date	01.10.2018	
Duration (months)	3	

Attachments

CV and publication list	CV_CarlosMelian.pdf CV_CeciliaAndreazzi.pdf
Further documents for scientific events or research visits	Form_confirmation_self_declaration_Signed.pdf
Other annexes	Research_Visit_Plan.pdf

1. Responsible applicant

Last name	Melian
First name	Carlos
Function (title)	Group Leader
Academic degree	Dr. / PhD
Date of birth	23.08.1969
Gender	male
Civil Status	Married
Language	German
Nationality	Spain
Correspondence address of application	Address of workplace

Home address

Address supplement	
Street, No.	Seestrasse 79
P.O. Box	
Postcode / Zipcode	6047
Place	Kastanienbaum
Country	Switzerland

Address of institute

Name of Institution 1 (e.g. laboratory) *	Eawag
Continuation 1 (e.g. inst / dept.)	
Continuation 2 (e.g. University)	
Street, No.	Seestrasse 79
Address supplement 1 (e.g. building)	
Address supplement 2 (e.g. office)	
P.O. Box	
Postcode / Zipcode	6047
Place	Kastanienbaum
State, canton, etc.	
Country	Switzerland

Communication

Secretariat line	
Switchboard	
Direct line	+41 58 765 22 08
Fax office	
Home telephone number	
Cellphone	+41 78 722 88 73
Website	http://www.eawag.ch/about/personen/homepages/melianca/index
E-mail address	carlos.melian@eawag.ch

2. Applicants' employment

Information on employment and function at the anticipated starting date of the grant

Name	Melian, Carlos
Employment at the anticipated starting date of the grant	permanent contract
Level of employment %	100
Function in the context of this grant application	Head of (e.g. institute, department, center, clinic)
Professorship	Assistant professor with tenure track
Doctorate (PhD)?	Yes
Date of doctorate (PhD)	01.02.2005
PhD supervisor	
Country of doctorate	Spain
Remarks	
Further employments	

3. Basic data I

Original title	Biodiversity Dynamics in Coevolutionary Metaecosystems (research visit)
Title in English	Biodiversity Dynamics in Coevolutionary Metaecosystems (research visit)
Requested starting date	01.10.2018
Duration (months)	3
Research field	Life sciences
Main discipline	30207 Ecology
Sub-discipline(s)	30206 Environmental Research

4. Basic data II

Summary	Background and rationale Evolutionary dynamics - the selection pressures affecting traits distributions - and ecological dynamics - the biotic and abiotic factors causing demographic changes on populations - can work out at similar spatio-temporal scales. Rapid evolutionary changes have been invoked as primary forces driving ecosystem function in species-rich meta-ecosystems. Yet, most theories developed to study the connection between evolutionary and ecological dynamics lack explicit reciprocal trait changes among many interacting species across heterogeneous landscapes. This gap might be limiting us to understand the effects of rapid reciprocal trait changes on meta-ecosystems functioning. Merging coevolutionary hypotheses and meta-ecosystem theory might improve predictions on biodiversity dynamics by taking into account the coevolutionary processes occurring at different spatio-temporal scales. For example, spatially variable biotic and abiotic selection and the migration of individuals among heterogeneous patches can generate complex selection mosaics leading to spatial variation in species traits. This leads to spatial variation in the structure and stability of ecological networks, which in turn may affect the coevolutionary process and the function of meta-ecosystems. Therefore, interactions between species might generate reciprocal selection pressures which can in turn affect the function of the meta-ecosystem. The coevolutionary dynamics consequence of species interactions that emerges over time varies with the fitness consequences of the ecological interaction (competitive, mutualistic, or antagonistic) and the architecture of the coevolving interactions. However, the connection between traits accounting for different interaction types and trait coevolution at small scales and the large-scale associations with biodiversity function and meta-ecosystem dynamics are not well understood. Thus, merging trait and space-based data containing many interacting species to analytical methods to explore coevolutionary dynamics among many interacting species at different spatial scales is critical to our understanding of the web of life. Objectives and aims During the research visit, we aim to integrate coevolutionary processes occurring in multiple ecological interactions into meta-ecosystem theory. The primary goals of our project are threefold: 1. Developing a "Coevolutionary meta-ecosystem model" to study the interdependent effects of the coevolution of traits, biodiversity dynamics and ecosystem function in an interaction network context. 2. Making explicit the connection between traits and demographic dynamics to assess how the speed of reciprocal trait change might be driving the dynamics of species interaction networks and ecosystem function, and 3. Merging meta-ecosystem dynamics and coevolutionary processes to explicitly link species trait distributions to the fitness and functional effects of the interactions. Methods
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We will extend the meta-ecosystem theory to study how reciprocal trait changes among many interacting species might affect biodiversity dynamics and ecosystem function. This overarching goal will be achieved by focusing on two main research questions:

Q1: How does reciprocal trait change influence biodiversity dynamics, species extinctions and ecosystem function in species-rich meta-ecosystems?

Q2: How do the spatial arrangement of ecosystems and coevolutionary dynamics interact to facilitate rapid evolutionary changes in meta-ecosystems?

During the research visit period we will develop the coevolutionary meta-ecosystem model to study feedbacks of reciprocal trait change in spatially explicit heterogeneous environments. Our model will be based in stochastic processes to include four interacting hierarchical levels (traits, population, ecosystem, and meta-ecosystem). The basic model will represent a meta-ecosystem with a network of P patches and S species per patch each containing initially N_i individuals per species i . Each species i in patch P undergoes trait dynamics driven by mating, gene flow (i.e., migration between patches), recombination, interaction selection (i.e., which drives the speed of reciprocal trait change) and environmental selection following an optimum for each species i . The distribution of traits of each species i will contain many phenotypes, each associated to a changing fitness value. We will track species trait coevolution and changes associated to ecosystem function using abundances and traits that drive the stability in each local patch. We will explicitly consider trait variance to calculate the fitness of a given trait in species-rich systems using the eco-evolutionary Gillespie algorithm. Using this model we will explore:

- 1) How different strengths of coevolutionary selection (i.e., reciprocal trait change, Q1), influence biodiversity dynamics in each local community and in the meta-ecosystem,
- 2) How different scenarios of reciprocal trait change (i.e., trait matching, trait mismatching, and exploitation barriers) modulate the number of coexisting species (Q1), and
- 3) How different spatial arrangements of interacting communities might facilitate or inhibit rapid changes in biodiversity and ecosystem function (Q2).

Expected results

This research visit will merge ecological and evolutionary processes across levels of biological organization, such as networks of species interactions and meta-ecosystems. The most novel expected results of this work are

- (1) how the speed of reciprocal trait changes at local scales might be responsible for large-scale associations between coevolution, biodiversity dynamics and meta-ecosystem functioning;
- (2) how reciprocal trait change governing interaction strength between species and biodiversity dynamics, might affect the stability of meta-ecosystems;
- (3) discussing predictions of the coevolutionary meta-ecosystem modeling scenarios in the face of current problems in epidemiology, agroecology and conservation biology, and
- (4) connecting micro-coevolutionary processes and macro-evolutionary patterns of biodiversity in meta-ecosystems.

Impact

We will produce an opinion, review or ideas-perspectives paper entitled “Biodiversity dynamics and ecosystem functioning in coevolutionary meta-ecosystems” for submission to *Trends in Ecology and Evolution*, *Ecology Letters* or *Biological Reviews*. We will first produce a working draft paper during the scientific event which we will develop further during the research visit following the objectives, aims and the method briefly outlined in this summary.

Keywords

Biodiversity dynamics
Metacommunity theory
Coevolution
Theoretical ecology
Biodiversity patterns

Language of correspondence

German

Financial administration

EAWAG Verwaltung zHv Frau Barbara Breu Rütli
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5. University or research institution

University

Eidg. Anstalt für Wasserversorgung - EAWAG
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Remarks

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6. Requested funding

Requested funding	Total (CHF)	Year 1
Total (CHF)	7'500	7'500

Research funds	Total (CHF)	Year 1
Travel & subsistence allowances	7'500	7'500
Total (CHF)	7'500	7'500
Total (%)	100%	100%

Details

Travel & subsistence allowances	Total (CHF)	Year 1
Andreazzi, Cecilia, Associate Researcher, Fundação Oswaldo Cruz, Rio de Janeiro, Brasil	7'500	7'500
Relation to research plan / Comments / Additions	Expenses per month (2 1/2 months) Housing 1750CHF x 2 + 875CHF = 4375CHF Food 900CHF x2 + 450CHF = 2250CHF Local transportation 350CHFx2 + 175CHF = 875CHF	
Total (CHF)	7'500	7'500
Total (%)	100%	100%

7. Timeframe of the scientific exchange

Starting date

15.10.2018

Place

Kastanienbaum, Switzerland

Duration (days)

75

8. Contents

General information
Aim and motivation

0

Organisation

0

Target audience
Expected number of people (total number of participants)

0

Remark

0

Active participants

Number of active participants from the academic field	0
Number of active participants from outside the academic field	0
Remark	0
Number of women	0
Number of PhD students	0
Number of students	0
Remark	0

9. General remarks on the project

Subject	Submitted in parallel with a scientific event
Communication	This research visit will be joined to a scientific event submitted in parallel.
Confidential	No